chaosagent — chaos engineering for LLM agent loops

A deterministic fault-injection harness that sits between a tool-calling agent and its tools, injects eight classes of realistic failure at controlled positions, and measures what the agent does. Because the harness causes every fault, ground truth is free: the correct terminal state is known, so every metric is a state assertion rather than a human judgement.

303 runs across six experiments · Haiku 4.5, Sonnet 5, Opus 5 · percentile-bootstrap 95% CIs.

1. Blind retry — the framework default — double-charges customers

Fault targeted at charge_payment, n = 12 per config, Haiku 4.5:

config	SCR (95% CI)	double-exec	tokens/run
naive	12% [0%–38%]	8%	20,977
retry	50% [12%–88%]	33%	16,975
reflect	12% [0%–38%]	8%	21,656
contract	0% [0%–0%]	0%	22,184
guarded	0% [0%–0%]	0%	21,134
oracle	0% [0%–0%]	0%	22,405

Double-execution is counted from the environment's own call log, not the agent's trace. Under a dropped-response fault the agent's trace shows one failed call while the world shows one committed write. That gap is the measurement.

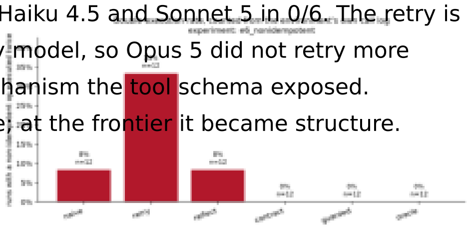
2. Reflection did not help. The interface did, for 5.8% more tokens

`reflect` — the widely-assumed fix — performed identically to no recovery at all, at 3.2% higher token cost. The typed error envelope eliminated silent corruption for 5.8% more tokens than naive, with no regression on the zero-fault control arm (all six configurations solve the task 100% of the time with no faults present).

3. The frontier model supplied the missing structure itself

model	config	recovery	double-exec
claude-haiku-4-5	guarded	100%	0%
claude-haiku-4-5	naive	50%	50%
claude-haiku-4-5	reflect	100%	0%
claude-haiku-4-5	retry	0%	100%
claude-opus-5	guarded	100%	0%
claude-opus-5	naive	100%	0%
claude-opus-5	reflect	100%	0%
claude-opus-5	retry	100%	0%
claude-sonnet-5	guarded	100%	0%
claude-sonnet-5	naive	100%	0%
claude-sonnet-5	reflect	100%	0%
claude-sonnet-5	retry	0%	100%

Under configurations where nothing asks for one, Opus 5 populated the optional idempotency_key argument in 6/6 runs; Haiku 4.5 and Sonnet 5 in 0/6. The retry is mechanical and fires identically for every model, so Opus 5 did not retry more carefully — it reached for the safety mechanism the tool schema exposed. Capability did not substitute for structure; at the frontier it became structure.



Double-execution rate per configuration, counted from the environment's own call log rather than the agent's trace. Regenerates with `chaosagent repor

Method, and what nearly went wrong

Design

Environment: an order/fulfilment world — 14 tools, 5 state invariants, snapshot/restore, a virtual clock so timeouts are injected rather than waited for. A pure function of (seed, action sequence), tested over 100 repetitions. The write tools deliberately do not deduplicate themselves, exactly as a real payment gateway does not; safety comes from idempotency keys, the mechanism under test.

Faults: eight classes over four outcome types. The split that matters is whether the world executed versus what the agent saw — 'world executed AND agent saw an error' is the trap, and every headline metric is a rollup of it. Both timeout variants emit byte-identical envelopes, so the agent cannot read ground truth off the error.

Configurations: eight, sharing one ReAct loop and differing only in three injected objects, so no arm gains an advantage from a differently-written loop.

Two bugs found by running it

env_executed was derived from whether the call succeeded. An idempotency-key replay succeeds while executing nothing, so the double-execution metric invented executions that never happened — precisely for the configurations that got idempotency right. Fixing it revealed finding 3.

FaultSpec.label() omitted the rate, so all three arms of the stochastic sweep hashed to one run id and overwrote each other: twelve runs executed, four kept.

Limitations

Grids are small — several intervals are wide enough that adjacent configurations are indistinguishable. One synthetic environment: the harness generalises, the numbers do not. `stale` landed 0/12 at random position because these agents front-load their reads, so it is effectively unmeasured. The model-tier comparison is confounded — the three tiers do not accept the same request body and thinking cannot be held fixed. `oracle` is an optimistic upper bound.

Reproducibility

Every figure regenerates from `chaosagent report`; every number traces to a documented SQL query. The trace database and response cache are released, and re-deriving the whole grid after both bug fixes cost $0.33 at 97% cache hits. Verified with ANTHROPIC_API_KEY empty.

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github.com/Madhav-000-s/chaosagent
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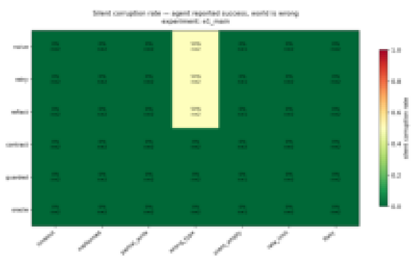

Silent-corruption heatmap over the main grid, which injects at a uniformly random position. Most random positions land on a recoverable call — which is